

Normalization

Databases - Computer Science

1. Functional dependencies

$X \rightarrow Y$ holds when any two tuples agreeing on X must agree on Y . Functional dependencies encode the rules of the domain and are the raw material of normalization. Armstrong axioms - reflexivity, augmentation, transitivity - derive every dependency implied by a set, and an attribute closure reveals whether a set is a superkey.

2. Update anomalies

A poorly factored table shows three symptoms. An insertion anomaly prevents recording a fact because unrelated information is missing. A deletion anomaly loses a fact as a side effect of removing another. An update anomaly requires the same fact to be changed in many rows, leaving inconsistency if any are missed.

3. The normal forms

First normal form requires atomic attributes. Second forbids partial dependency of a non-key attribute on part of a composite key. Third forbids transitive dependency of one non-key attribute on another. Boyce-Codd tightens third: every non-trivial determinant must be a superkey.

4. Denormalising deliberately

Normalization optimises for correctness under update; read-heavy workloads sometimes pay for that in join cost. A measured, documented denormalization - a materialised aggregate, a duplicated lookup column - is legitimate engineering. An undocumented duplicate is a bug waiting to happen.